

MicroserviceBase

v. 0.1.1

TODO

02.02.2023

Contents

1	Introduction	1
2	Description	2
2.1	ToDo	2
2.1.1	ToDo ToDo	2
3	__init__.py	3
3.1	Function: readPlist	3
3.2	Function: writePlist	3
3.3	Function: readPlistFromString	3
3.4	Function: writePlistToString	3
3.5	Function: is_stream_binary_plist	3
3.6	Class: Uid	3
3.6.1	Method: parse	5
3.6.2	Method: reset	5
3.6.3	Method: readRoot	5
3.6.4	Method: setCurrentOffsetToObjectNumber	5
3.6.5	Method: beginOffsetProtection	5
3.6.6	Method: endOffsetProtection	5
3.6.7	Method: readObject	5
3.6.8	Method: readContents	5
3.6.9	Method: readInteger	5
3.6.10	Method: readReal	5
3.6.11	Method: readRefs	5
3.6.12	Method: readArray	5
3.6.13	Method: readDict	5
3.6.14	Method: readAsciiString	5
3.6.15	Method: readUnicode	5
3.6.16	Method: readDate	5
3.6.17	Method: readData	5
3.6.18	Method: readUid	5
3.6.19	Method: getSizedInteger	5
3.7	Class: BoolWrapper	5
3.8	Class: FloatWrapper	5
3.9	Class: StringWrapper	6
3.9.1	Method: encodingMarker	6
3.10	Class: PlistWriter	6

3.10.1	Method: reset	6
3.10.2	Method: positionOfObjectReference	6
3.10.3	Method: endRecursionProtection	6
3.10.4	Method: wrapRoot	6
3.10.5	Method: incrementByteCount	6
3.10.6	Method: computeOffsets	6
3.10.7	Method: writeObjectReference	6
3.10.8	Method: binaryInt	7
3.10.9	Method: intSize	7
4	badge.py	8
4.1	Function: badge_disk_icon	8
5	colors.py	9
5.1	Function: isAColor	9
5.2	Function: parseColor	9
5.3	Class: Color	9
5.3.1	Method: to_rgb	9
5.4	Class: RGB	9
5.4.1	Method: to_rgb	9
5.5	Class: HSL	9
5.5.1	Method: to_rgb	9
5.6	Class: HWB	9
5.6.1	Method: to_rgb	10
5.7	Class: CMYK	10
5.7.1	Method: to_rgb	10
5.8	Class: Gray	10
5.8.1	Method: to_rgb	10
5.9	Class: ColorParser	10
5.9.1	Method: skipws	11
5.9.2	Method: expect	11
5.9.3	Method: expectEnd	11
5.9.4	Method: getToken	11
5.9.5	Method: parseNumber	11
5.9.6	Method: parseColor	11
5.9.7	Method: parseRGB	11
5.9.8	Method: parseHSL	11
5.9.9	Method: parseHWB	11
5.9.10	Method: parseCMYK	11
5.9.11	Method: parseGray	11
5.9.12	Method: parseValue	11
5.9.13	Method: parseAngle	11
6	core.py	12
6.1	Function: build_dmg	12
6.2	Class: DMGError	12

7	buddy.py	13
7.1	Class: BuddyError	13
7.2	Class: Block	13
7.2.1	Method: close	13
7.2.2	Method: flush	13
7.2.3	Method: invalidate	13
7.2.4	Method: zero_fill	13
7.2.5	Method: tell	13
7.2.6	Method: seek	13
7.2.7	Method: read	13
7.2.8	Method: write	13
7.2.9	Method: insert	13
7.2.10	Method: delete	13
7.3	Class: Allocator	13
7.3.1	Method: open	14
7.3.2	Method: close	14
7.3.3	Method: flush	14
7.3.4	Method: read	14
7.3.5	Method: allocate	14
7.3.6	Method: iterkeys	14
7.3.7	Method: keys	14
8	store.py	15
8.1	Class: ILocCodec	15
8.1.1	Method: encode	15
8.1.2	Method: decode	15
8.2	Class: PlistCodec	15
8.2.1	Method: encode	15
8.2.2	Method: decode	15
8.3	Class: BookmarkCodec	15
8.3.1	Method: encode	15
8.3.2	Method: decode	15
8.4	Class: DSStoreEntry	15
9	alias.py	18
9.1	Function: encode_utf8	18
9.2	Function: decode_utf8	18
9.3	Class: AppleShareInfo	18
9.4	Class: VolumeInfo	18
9.4.1	Method: filesystem_type	18
9.5	Class: TargetInfo	18
9.6	Class: Alias	18
9.6.1	Method: from_bytes	19
10	bookmark.py	20
10.1	Function: iteritems	20
10.2	Class: Data	20

10.3 Class: URL	20
10.3.1 Method: absolute	20
10.3.2 Method: from_bytes	20
11 osx.py	21
11.1 Function: getattrlist	21
11.2 Function: fgetattrlist	21
11.3 Function: statfs	21
11.4 Function: fstatfs	21
11.5 Class: attrlist	21
11.6 Class: attribute_set_t	21
11.7 Class: fsobj_id_t	21
11.8 Class: timespec	21
11.9 Class: attrreference_t	22
11.10Class: fsid_t	22
11.11Class: guid_t	22
11.12Class: kauth_ace	22
11.13Class: kauth_acl	22
11.14Class: kauth_filesec	22
11.15Class: diskextent	23
11.16Class: Point	23
11.17Class: Rect	23
11.18Class: FileInfo	23
11.19Class: FolderInfo	23
11.20Class: FinderInfo	23
11.21Class: vol_capabilities_attr_t	23
11.22Class: vol_attributes_attr_t	24
11.23Class: struct_statfs	24
12 utils.py	25
12.1 Class: UTC	25
12.1.1 Method: utcoffset	25
12.1.2 Method: dst	25
12.1.3 Method: tzname	25
13 ServiceBase.py	26
13.1 Class: ResultType	26
13.2 Class: ResponseMessage	26
13.2.1 Method: get_json	26
13.2.2 Method: get_data	26
13.3 Class: ServiceBase	26
13.3.1 Method: parse_arguments	27
13.3.2 Method: parse_spec_arguments	27
13.3.3 Method: close	27
13.3.4 Method: create_request_data	27
13.3.5 Method: connect_broker	28
13.3.6 Method: serve	28

13.3.7 Method: <code>register_service</code>	28
13.3.8 Method: <code>unregister_service</code>	28
13.3.9 Method: <code>request_service</code>	28
13.3.10 Method: <code>get_svc_api_methods_dict</code>	29
13.3.11 Method: <code>get_svc_api_methods_info_dict</code>	29
13.3.12 Method: <code>parse_docstring</code>	29
13.3.13 Method: <code>svc_api_get_version</code>	29
13.3.14 Method: <code>svc_api_get_gui_files</code>	29
13.3.15 Method: <code>is_specific_request</code>	29
13.3.16 Method: <code>on_specific_request</code>	30
13.3.17 Method: <code>on_request</code>	30
14 ServiceRegistry.py	31
14.1 Function: <code>signal_handler</code>	31
14.2 Class: <code>ServiceRegistry</code>	31
14.2.1 Method: <code>receive_services_information</code>	31
14.2.2 Method: <code>handle_update</code>	31
14.2.3 Method: <code>notify_updates</code>	32
14.2.4 Method: <code>svc_api_get_services_info</code>	32
14.2.5 Method: <code>svc_api_get_realtime_update_exchange</code>	32
14.2.6 Method: <code>svc_api_update_alias_conf</code>	32
14.2.7 Method: <code>svc_api_get_alias_conf</code>	33
14.2.8 Method: <code>is_specific_request</code>	33
14.2.9 Method: <code>on_specific_request</code>	33
15 Appendix	34
16 History	35

Chapter 1

Introduction

TODO: Add introduction of **MicroserviceBase**.

The sources of **MicroserviceBase** are available in [GitHub](#).

Informations about how to install the **MicroserviceBase** can be found in the [README](#).

Chapter 2

Description

2.1 ToDo

Description

2.1.1 ToDo ToDo

Description

ToDo ToDo ToDo

Description

Chapter 3

`__init__.py`

biplist -- a library for reading and writing binary property list files.

Binary Property List (plist) files provide a faster and smaller serialization format for property lists on OS X. This is a library for generating binary plists which can be read by OS X, iOS, or other clients.

The API models the plistlib API, and will call through to plistlib when XML serialization or deserialization is required.

To generate plists with UID values, wrap the values with the `Uid` object. The value must be an int.

To generate plists with `NSData`/`CFData` values, wrap the values with the `Data` object. The value must be a string.

Date values can only be `datetime.datetime` objects.

The exceptions `InvalidPlistException` and `NotBinaryPlistException` may be thrown to indicate that the data cannot be serialized or deserialized as a binary plist.

Plist generation example:

```
from biplist import * from datetime import datetime plist = {'aKey':'aValue', '0':1.322, 'now':datetime.now(),
'list':[1,2,3], 'tuple':('a','b','c')} try: writePlist(plist, "example.plist") except (InvalidPlistException,
NotBinaryPlistException), e: print "Something bad happened:", e
```

Plist parsing example:

```
from biplist import * try: plist = readPlist("example.plist") print plist except (InvalidPlistException,
NotBinaryPlistException), e: print "Not a plist:", e
```

3.1 Function: `readPlist`

Raises `NotBinaryPlistException`, `InvalidPlistException` `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---wrapdataobject =====`

3.2 Function: `writePlist`

3.3 Function: `readPlistFromString`

3.4 Function: `writePlistToString`

3.5 Function: `is_stream_binary_plist`

3.6 Class: `Uid`

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.bplist.__init__
↳ import Uid
```

Wrapper around integers for representing UID values. This is used in keyed archiving. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-bplist__init__data =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.bplist.__init__
↳ import Data
```

Wrapper around bytes to distinguish Data values. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-bplist__init__invalidplistexception =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.bplist.__init__
↳ import InvalidPlistException
```

Raised when the plist is incorrectly formatted. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-bplist__init__notbinaryplistexception =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.bplist.__init__
↳ import NotBinaryPlistException
```

Raised when a binary plist was expected but not encountered. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-bplist__init__plistreader =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.bplist.__init__
↳ import PlistReader
```

- 3.6.1 Method: parse
- 3.6.2 Method: reset
- 3.6.3 Method: readRoot
- 3.6.4 Method: setCurrentOffsetToObjectNumber
- 3.6.5 Method: beginOffsetProtection
- 3.6.6 Method: endOffsetProtection
- 3.6.7 Method: readObject
- 3.6.8 Method: readContents
- 3.6.9 Method: readInteger
- 3.6.10 Method: readReal
- 3.6.11 Method: readRefs
- 3.6.12 Method: readArray
- 3.6.13 Method: readDict
- 3.6.14 Method: readAsciiString
- 3.6.15 Method: readUnicode
- 3.6.16 Method: readDate
- 3.6.17 Method: readData
- 3.6.18 Method: readUid
- 3.6.19 Method: getSizedInteger

Numbers of 8 bytes are signed integers when they refer to numbers, but unsigned otherwise. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---hashablewrapper =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import HashableWrapper
```

3.7 Class: BoolWrapper

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import BoolWrapper
```

3.8 Class: FloatWrapper

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import FloatWrapper
```

3.9 Class: StringWrapper

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import StringWrapper
```

3.9.1 Method: encodingMarker

3.10 Class: PlistWriter

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import PlistWriter
```

3.10.1 Method: reset

3.10.2 Method: positionOfObjectReference

If the given object has been written already, return its position in the offset table. Otherwise, return None.
microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeroot -----

Strategy is: - write header - wrap root object so everything is hashable - compute size of objects which will be written - need to do this in order to know how large the object refs will be in the list/dict/set reference lists - write objects - keep objects in writtenReferences - keep positions of object references in referencePositions - write object references with the length computed previously - computer object reference length - write object reference positions - write trailer microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-beginrecursionprotection -----

3.10.3 Method: endRecursionProtection

3.10.4 Method: wrapRoot

3.10.5 Method: incrementByteCount

3.10.6 Method: computeOffsets

3.10.7 Method: writeObjectReference

Tries to write an object reference, adding it to the references table. Does not write the actual object bytes or set the reference position. Returns a tuple of whether the object was a new reference (True if it was, False if it already was in the reference table) and the new output. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeobject -----

Serializes the given object to the output. Returns output. If setReferencePosition is True, will set the position the object was written. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeoffsettable -----

Writes all of the object reference offsets. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-binaryreal` -----

3.10.8 Method: `binaryInt`

3.10.9 Method: `intSize`

Returns the number of bytes necessary to store the given integer. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-realsize` -----

Chapter 4

badge.py

4.1 Function: `badge_disk_icon`

Chapter 5

colors.py

5.1 Function: isAColor

5.2 Function: parseColor

5.3 Class: Color

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import Color
```

5.3.1 Method: to_rgb

5.4 Class: RGB

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import RGB
```

5.4.1 Method: to_rgb

5.5 Class: HSL

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import HSL
```

5.5.1 Method: to_rgb

5.6 Class: HWB

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import HWB
```

5.6.1 Method: to_rgb

5.7 Class: CMYK

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import CMYK
```

5.7.1 Method: to_rgb

5.8 Class: Gray

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import Gray
```

5.8.1 Method: to_rgb

5.9 Class: ColorParser

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import ColorParser
```


- 5.9.1 Method: skipws
- 5.9.2 Method: expect
- 5.9.3 Method: expectEnd
- 5.9.4 Method: getToken
- 5.9.5 Method: parseNumber
- 5.9.6 Method: parseColor
- 5.9.7 Method: parseRGB
- 5.9.8 Method: parseHSL
- 5.9.9 Method: parseHWB
- 5.9.10 Method: parseCMYK
- 5.9.11 Method: parseGray
- 5.9.12 Method: parseValue
- 5.9.13 Method: parseAngle

Chapter 6

core.py

6.1 Function: build_dmg

6.2 Class: DMGError

Imported by:

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.core  
↳ import DMGError
```

Chapter 7

buddy.py

7.1 Class: BuddyError

Imported by:

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.buddy  
↪ import BuddyError
```

7.2 Class: Block

Imported by:

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.buddy  
↪ import Block
```

7.2.1 Method: close

7.2.2 Method: flush

7.2.3 Method: invalidate

7.2.4 Method: zero_fill

7.2.5 Method: tell

7.2.6 Method: seek

7.2.7 Method: read

7.2.8 Method: write

7.2.9 Method: insert

7.2.10 Method: delete

7.3 Class: Allocator

Imported by:

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds-store.buddy
↳ import Allocator

```

7.3.1 Method: open

7.3.2 Method: close

7.3.3 Method: flush

7.3.4 Method: read

Read data at 'offset', or raise an exception. 'size_or_format' may either be a byte count, in which case we return raw data, or a format string for 'struct.unpack', in which case we work out the size and unpack the data before returning it. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-write -----

Write data at 'offset', or raise an exception. 'data_or_format' may either be the data to write, or a format string for 'struct.pack', in which case we pack the additional arguments and write the resulting data. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-get-block -----

7.3.5 Method: allocate

Allocate or reallocate a block such that it has space for at least 'bytes' bytes. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-release -----

7.3.6 Method: iterkeys

7.3.7 Method: keys

Chapter 8

store.py

8.1 Class: ILocCodec

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import ILocCodec
```

8.1.1 Method: encode

8.1.2 Method: decode

8.2 Class: PlistCodec

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import PlistCodec
```

8.2.1 Method: encode

8.2.2 Method: decode

8.3 Class: BookmarkCodec

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import BookmarkCodec
```

8.3.1 Method: encode

8.3.2 Method: decode

8.4 Class: DSStoreEntry

Imported by:

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import DSStoreEntry

```

Holds the data from an entry in a `.DS_Store` file. Note that this is not meant to represent the entry itself--i.e. if you change the type or value, your changes will *not* be reflected in the underlying file.

If you want to make a change, you should either use the `DSStore` object's `DSStore.insert` method (which will replace a key if it already exists), or the mapping access mode for `DSStore` (often simpler anyway). `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-read` -----

Read a `.DS_Store` entry from the containing Block `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-byte-length` -----

Compute the length of this entry, in bytes `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-write` -----

Write this entry to the specified Block `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore` =====

Imported by:

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import DSStore

```

Python interface to a `.DS_Store` file. Works by manipulating the file on the disk--so this code will work with `.DS_Store` files for *very* large directories.

A `DSStore` object can be used as if it was a mapping, e.g.:

```
d['foobar.dat']['Iloc']
```

will fetch the "Iloc" record for "foobar.dat", or raise `KeyError` if there is no such record. If used in this manner, the `DSStore` object will return (type, value) tuples, unless the type is "blob" and the module knows how to decode it.

Currently, we know how to decode "Iloc", "bwsp", "lsvp", "lsvP" and "icvp" blobs. "Iloc" decodes to an (x, y) tuple, while the others are all decoded using `biplist`.

Assignment also works, e.g.:

```
d['foobar.dat']['note'] = ('ustr', u'Hello World!')
```

as does deletion with `del`:

```
del d['foobar.dat']['note']
```

This is usually going to be the most convenient interface, though occasionally (for instance when creating a new `.DS_Store` file) you may wish to drop down to using `DSStoreEntry` objects directly. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-open` -----

Open a `.DS_Store` file; pass either a Python file object, or a filename in the `file_or_name` argument and a file access mode in the `mode` argument. If you are creating a new file using the "w" or "w+" modes, you may also specify a list of entries with which to initialise the file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-flush` -----

Flush any dirty data back to the file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-close` -----

Flush dirty data and close the underlying file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-insert` -----

Insert entry (which should be a `DSStoreEntry`) into the B-Tree. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-delete` -----

Delete an item, identified by `filename` and `code` from the B-Tree. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-find` -----

Returns a generator that will iterate over matching entries in the B-Tree.

Chapter 9

alias.py

9.1 Function: encode_utf8

9.2 Function: decode_utf8

9.3 Class: AppleShareInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import AppleShareInfo
```

9.4 Class: VolumeInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import VolumeInfo
```

9.4.1 Method: filesystem_type

9.5 Class: TargetInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import TargetInfo
```

9.6 Class: Alias

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import Alias
```


9.6.1 Method: from_bytes

Construct an `Alias` object given binary `Alias` data. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-alias-alias-for-file` -----

Create an `Alias` that points at the specified file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-alias-alias-to-bytes` -----

Returns the binary representation for this `Alias`.

Chapter 10

bookmark.py

10.1 Function: iteritems

10.2 Class: Data

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import Data
```

10.3 Class: URL

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import URL
```

10.3.1 Method: absolute

Return an absolute URL. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark =====

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import Bookmark
```

10.3.2 Method: from_bytes

Create a `Bookmark` given byte data. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-get -----

Lookup the value for a given key, returning a default if not present. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-to-bytes -----

Convert this `Bookmark` to a byte representation. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-for-file -----

Construct a `Bookmark` for a given file.

Chapter 11

osx.py

11.1 Function: getattrlist

11.2 Function: fgetattrlist

11.3 Function: statfs

11.4 Function: fstatfs

11.5 Class: attrlist

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attrlist
```

11.6 Class: attribute_set_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attribute_set_t
```

11.7 Class: fsobj_id_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import fsobj_id_t
```

11.8 Class: timespec

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import timespec
```

11.9 Class: attrreference_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attrreference_t
```

11.10 Class: fsid_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import fsid_t
```

11.11 Class: guid_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import guid_t
```

11.12 Class: kauth_ace

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_ace
```

11.13 Class: kauth_acl

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_acl
```

11.14 Class: kauth_filesec

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_filesec
```

11.15 Class: diskextent

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import diskextent
```

11.16 Class: Point

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import Point
```

11.17 Class: Rect

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import Rect
```

11.18 Class: FileInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FileInfo
```

11.19 Class: FolderInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FolderInfo
```

11.20 Class: FinderInfo

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FinderInfo
```

11.21 Class: vol_capabilities_attr_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import vol_capabilities_attr_t
```

11.22 Class: vol_attributes_attr_t

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import vol_attributes_attr_t
```

11.23 Class: struct_statfs

Imported by:

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import struct_statfs
```

Chapter 12

utils.py

12.1 Class: UTC

Imported by:

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.utils  
↪ import UTC
```

12.1.1 Method: utcoffset

12.1.2 Method: dst

12.1.3 Method: tzname

Chapter 13

ServiceBase.py

13.1 Class: ResultType

Imported by:

```
from MicroserviceBase.ServiceRegistry.ServiceBase import ResultType
```

Result Types.

13.2 Class: ResponseMessage

Imported by:

```
from MicroserviceBase.ServiceRegistry.ServiceBase import ResponseMessage
```

Response message class

13.2.1 Method: get_json

Convert the response message to JSON format.

Returns:

/ Type: str /

The response message in JSON format.

13.2.2 Method: get_data

Retrieve the result data as a string.

Returns:

/ Type: str /

The result data as a string.

13.3 Class: ServiceBase

Imported by:

```
from MicroserviceBase.ServiceRegistry.ServiceBase import ServiceBase
```


13.3.1 Method: `parse_arguments`

Parse basic service arguments from the command line.

Arguments:

- `cmd_args`
/ *Condition*: required / *Type*: list /
Command-line arguments to be parsed.

Returns:

/ *Type*: dict /
A dictionary containing the parsed arguments.

13.3.2 Method: `parse_spec_arguments`

Parse specific arguments for each customized service.

This method should be overridden by subclasses to handle service-specific arguments.

Arguments:

- `cmd_args`
/ *Condition*: required / *Type*: list /
Command-line arguments to be parsed.

Returns:

/ *Type*: dict /
A dictionary containing the parsed specific arguments.

13.3.3 Method: `close`

Close the service connection.

Returns:

(no returns)

13.3.4 Method: `create_request_data`

Create request data for a given method name and arguments.

Arguments:

- `method_name`
/ *Condition*: required / *Type*: str /
The name of the method for which the request is being created.
- `args`
/ *Condition*: required / *Type*: list /
The list of arguments for the method.

Returns:

/ *Type*: dict /
A dictionary containing the method name and arguments.

13.3.5 Method: connect_broker

Establish a connection to the broker.

Arguments:

- `**kwargs`
/ *Condition*: optional / *Type*: dict /
Additional keyword arguments for broker connection parameters.

Returns:

(no returns)

13.3.6 Method: serve

Call to start service serving.

Returns:

(no returns)

13.3.7 Method: register_service

Register a service to the ServiceRegistry.

Returns:

(no returns)

13.3.8 Method: unregister_service

Unregister a service from the ServiceRegistry.

Returns:

(no returns)

13.3.9 Method: request_service

Send a service request to a specific exchange with a given routing key.

Arguments:

- `request_data`
/ *Condition*: required / *Type*: dict /
The data for the service request.
- `exchange_name`
/ *Condition*: required / *Type*: str /
The name of the exchange to send the request to.
- `routing_key`
/ *Condition*: required / *Type*: str /
The routing key for the request.

Returns:

(no returns)

13.3.10 Method: `get_svc_api_methods_dict`

Retrieve all service API methods provided by the service (methods starting with the prefix 'svc_api').

Returns:

/ Type: dict /

A dictionary containing the names and references of all service API methods.

13.3.11 Method: `get_svc_api_methods_info_dict`

Retrieve information for all service APIs from the docstrings of the methods.

Arguments:

- `methods_dict`

/ Condition: required / Type: dict /

A dictionary containing the names and references of all service API methods.

Returns:

/ Type: dict /

A dictionary containing the information of all service APIs extracted from their docstrings.

13.3.12 Method: `parse_docstring`

Parse function's docstring to get arguments and return information.

Arguments:

- `docstring`

/ Condition: required / Type: str /

Function's docstring.

Returns:

/ Type: str /

Payloads of the waiting signal if received.

13.3.13 Method: `svc_api_get_version`

Get the service version.

Returns:

/ Type: str /

Version of the service.

13.3.14 Method: `svc_api_get_gui_files`

13.3.15 Method: `is_specific_request`

Check if the request is a specific request.

Arguments:

- `request`

/ Condition: required / Type: object /

The request object to be checked.

Returns:

/ Type: bool /

True if the request is a specific request, otherwise False.

13.3.16 Method: on_specific_request

Handle the event when a specific request is received.

Arguments:

- `ch`
/ Condition: required / Type: pika.channel.Channel /
The channel object from the pika library.
- `method`
/ Condition: required / Type: pika.spec.Basic.Deliver /
The method object containing delivery information from the pika library.
- `props`
/ Condition: required / Type: pika.spec.BasicProperties /
The properties of the message from the pika library.
- `body`
/ Condition: required / Type: dict /
The body of the message as a dictionary.

Returns:

(no returns)

13.3.17 Method: on_request

Handle an incoming request.

Arguments:

- `ch`
/ Condition: required / Type: pika.channel.Channel /
The channel object from the pika library.
- `method`
/ Condition: required / Type: pika.spec.Basic.Deliver /
The method object containing delivery information from the pika library.
- `props`
/ Condition: required / Type: pika.spec.BasicProperties /
The properties of the message from the pika library.
- `body`
/ Condition: required / Type: bytes /
The body of the message as bytes.

Returns:

(no returns)

Chapter 14

ServiceRegistry.py

14.1 Function: signal_handler

Handle signals from the operating system.

Arguments:

- `sig`
/ *Condition*: required / *Type*: int /
The signal number received from the OS.
- `frame`
/ *Condition*: required / *Type*: frame object /
The current stack frame.
- `obj`
/ *Condition*: required / *Type*: object /
The object that is handling the signal.

Returns:

(no returns)

14.2 Class: ServiceRegistry

Imported by:

```
from MicroserviceBase.ServiceRegistry.ServiceRegistry import ServiceRegistry
```

Manage information for all services connected to the broker it is connected to.

This class handles the registration, updates, and retrieval of service information, ensuring that all services interacting with the broker are properly managed and monitored.

14.2.1 Method: receive_services_information

Run in a thread to listen for any changes from the services.

Returns:

(no returns)

14.2.2 Method: handle_update

Handle the event when service information is updated.

Arguments:

- `ch`
/ *Condition*: required / *Type*: `pika.channel.Channel` /
The channel object from the pika library.
- `method`
/ *Condition*: required / *Type*: `pika.spec.Basic.Deliver` /
The method object containing delivery information from the pika library.
- `properties`
/ *Condition*: required / *Type*: `pika.spec.BasicProperties` /
The properties of the message from the pika library.
- `body`
/ *Condition*: required / *Type*: `bytes` /
The body of the message as bytes.

Returns:

(no returns)

14.2.3 Method: `notify_updates`

Notify updates to the realtime update channel.

Returns:

(no returns)

14.2.4 Method: `svc_api_get_services_info`

Retrieve information of all services connected to the broker that the Service Registry is connected to.

Returns:

/ *Type*: `dict` /

A dictionary containing information of all connected services.

14.2.5 Method: `svc_api_get_realtime_update_exchange`

Retrieve the exchange name of the realtime update exchange.

Returns:

/ *Type*: `str` /

The name of the realtime update exchange.

14.2.6 Method: `svc_api_update_alias_conf`

Update the alias configuration information.

Arguments:

- `alias_string`
/ *Condition*: required / *Type*: `str` /
The alias configuration string to be updated.

Returns:

(no returns)

14.2.7 Method: `svc_api_get_alias_conf`

Retrieve the alias configuration string in JSON format.

Returns:

/ Type: str /

The alias configuration string in JSON format.

14.2.8 Method: `is_specific_request`

Check if the request is a specific request.

Arguments:

- `request`

/ Condition: required / Type: object /

The request object to be checked.

Returns:

/ Type: bool /

True if the request is a specific request, otherwise False.

14.2.9 Method: `on_specific_request`

Handle the event when a specific request is received.

Arguments:

- `ch`

/ Condition: required / Type: pika.channel.Channel /

The channel object from the pika library.

- `method`

/ Condition: required / Type: pika.spec.Basic.Deliver /

The method object containing delivery information from the pika library.

- `props`

/ Condition: required / Type: pika.spec.BasicProperties /

The properties of the message from the pika library.

- `body`

/ Condition: required / Type: dict /

The body of the message as a dictionary.

Returns:

(no returns)

Chapter 15

Appendix

About this package:

Table 15.1: Package setup

Setup parameter	Value
Name	MicroserviceBase
Version	0.1.1
Date	02.02.2023
Description	TODO
Package URL	python-microservice-base
Author	TODO
Email	TODO
Language	Programming Language :: Python :: 3
License	License :: OSI Approved :: Apache Software License
OS	Operating System :: OS Independent
Python required	>=3.0
Development status	Development Status :: 4 - Beta
Intended audience	Intended Audience :: Developers
Topic	Topic :: Software Development

Chapter 16

History

0.1.0	02/2023
<i>Initial version</i>	